United International University (UIU)
Dept. of Electrical and Electronic Engineering (EEE)
Course No. EEE 312
Course Title: Digital Signal Processing Lab

Lab Sheet 4
Frequency Domain Analysis using MATLAB

Prerequisite

Before starting this section, students should know..

1. A fundamental knowledge on MATLAB
2. A theoretical knowledge on z-transform and DTFT

Outcomes

After finishing this lab students should be able to.....

1. understand frequency domain analysis of DT signal using MATLAB.
2. understand stability of DT system using MATLAB.

Outlines of this Lab

1. Lab Session 4.1: Z-transform
2. Lab Session 4.2: Inverse Z-transform
3. Lab Session 4.3: Pole zero representation
4. Lab Session 4.4: Frequency response
5. Lab Session 4.5: Stability and Causality using z-Transform
6. Lab Session 4.6: Introduction to DTFT
7. Lab Session 4.7: In Lab Evaluation
8. Home work

Lab Session 4.1: Z-transform

The Z-transform converts a discrete time domain signal, which is a sequence of real numbers, into a complex frequency domain representation.

The Z-transform of a discrete-time signal *x(n)* is the function *X(z)* defined as

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

where *n* is an integer and *z* is, in general, a complex number.

First OBE version, prepared by: B.K.M. Mizanur Rahman, Assistant Professor, Department of EEE, UIU

Z-transform using ztrans() function

Example 4.1: Find the z-transform of $x(\text{n}) = \text{a}^{\text{n}} u(\text{n})$

MATLAB Code:

```
syms a n x
x = a^n;
ztrans(x)
```

Output:

ans =

-z/(a - z)

Example 4.2: Find the z-transform of $x(\text{n}) = n(0.5)^{\text{n}} u(\text{n})$

MATLAB Code:

```
syms n x
x = n*(0.5)^n;
ztrans(x)
```

Output:

ans =

(2*z)/(2*z - 1)^2

(Note that ztrans is used in a causal case only.)

Lab Session 4.2: Inverse Z-transform

The inverse Z-transform is the conversion of Z-domain signal into time domain signal. The inversion can be done by Cauchy's Integral theorem, long division process, partial fraction expansion etc.

Inverse Z-transform using iztrans() function

Example 4.3: Find the inverse z-transform of $\text{X(z)} = \frac{\text{z}}{(\text{z}-0.2)(\text{z}-0.6)}$

MATLAB Code:

```
syms z y
y=z/((z-0.2)*(z-0.6));
iztrans(y)
```

Output:

ans =

(5*(3/5)^n)/2 - (5*(1/5)^n)/2

(Note that iztrans always results in a causal case only.)

Lab Session 4.3: Pole zero representation

The poles are those values for which the system transfer function becomes infinite. These are the roots of the denominator of a transfer function.

The zeros are those values for which the system transfer function becomes zero. These are the roots of the numerator of a transfer function.

The pole zero plot of a system provides information about the system behavior.

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Pole Zero representation using zplane() function

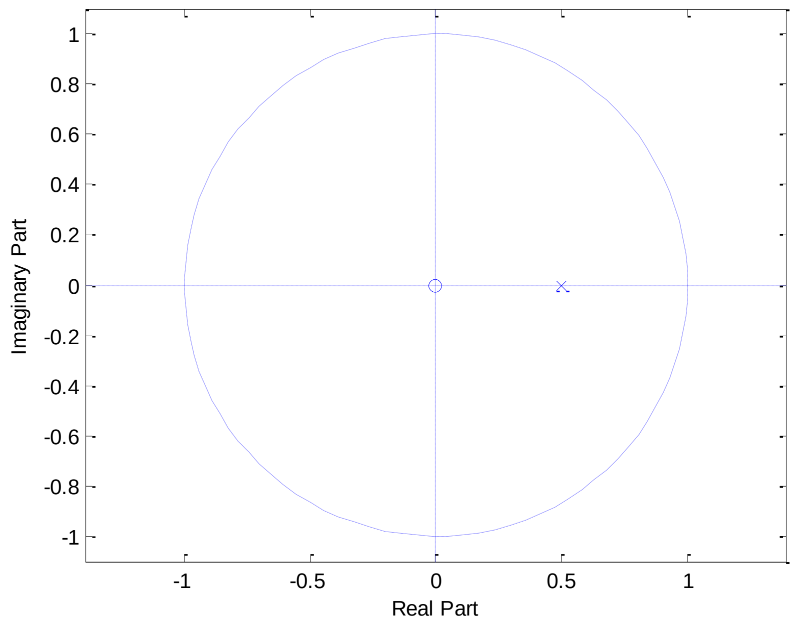
With the function **zplane()**, the pole zero plot can be obtained from which we can get the information of system behavior.

Example 4.4: For a transfer function $H(z) = \dfrac{1}{1 - 0.5z^{-1}}$, give the pole zero representation.

MATLAB Code:

```
b=[1];
a=[1 -0.5];
zplane(b,a)
```

Output:



Lab Session 4.4: Frequency response

If the ROC of $H(z)$ includes a unit circle ($z = e^{j\omega}$), then we can evaluate $H(z)$ on the unit circle, resulting in a frequency response function or transfer function $H(e^{j\omega})$.

The general expression for H(z) is ,

$$H(z) = b_0\ z^{N-M}\ \frac{\prod_{\ell=1}^{N}(z - z_\ell)}{\prod_{k=1}^{N}(z - p_k)}$$

Putting $z = e^{j\omega}$

$$H(e^{j\omega}) = b_0\ e^{j(N-M)\omega}\ \frac{\prod_1^M (e^{j\omega} - z_\ell)}{\prod_1^N (e^{j\omega} - p_k)}$$

therefore,

$$|H(e^{j\omega})| = |b_0| \frac{|e^{j\omega} - z_1| \cdots |e^{j\omega} - z_M|}{|e^{j\omega} - p_1| \cdots |e^{j\omega} - p_N|}$$

It can be interpreted as a product of the lengths of vectors from zeros to the unit circle divided by the lengths of vectors from poles to the unit circle and scaled by |b0|. Similarly, the phase response function

$$\angle H(e^{j\omega}) = \underbrace{[0 \text{ or } \pi]}_{\text{Constant}} + \underbrace{[(N-M)\,\omega]}_{\text{Linear}} + \underbrace{\sum_{1}^{M} \angle(e^{j\omega} - z_k) - \sum_{1}^{N} \angle(e^{j\omega} - p_k)}_{\text{Nonlinear}}$$

can be interpreted as a sum of a constant factor, a linear-phase factor, and a nonlinear-phase factor (angles from the "zero vectors" minus the sum of angles from the "pole vectors").

freqz() function can be used in MATLAB to plot frequency response.

Example 4.5:

Given a causal system

$$y(n) = 0.9y(n-1) + x(n)$$

a. Determine $H(z)$ and sketch its pole-zero plot.
b. Plot $|H(e^{j\omega})|$ and $\angle H(e^{j\omega})$.
c. Determine the impulse response $h(n)$.

Solution:

(a) The difference equation can be put in the form

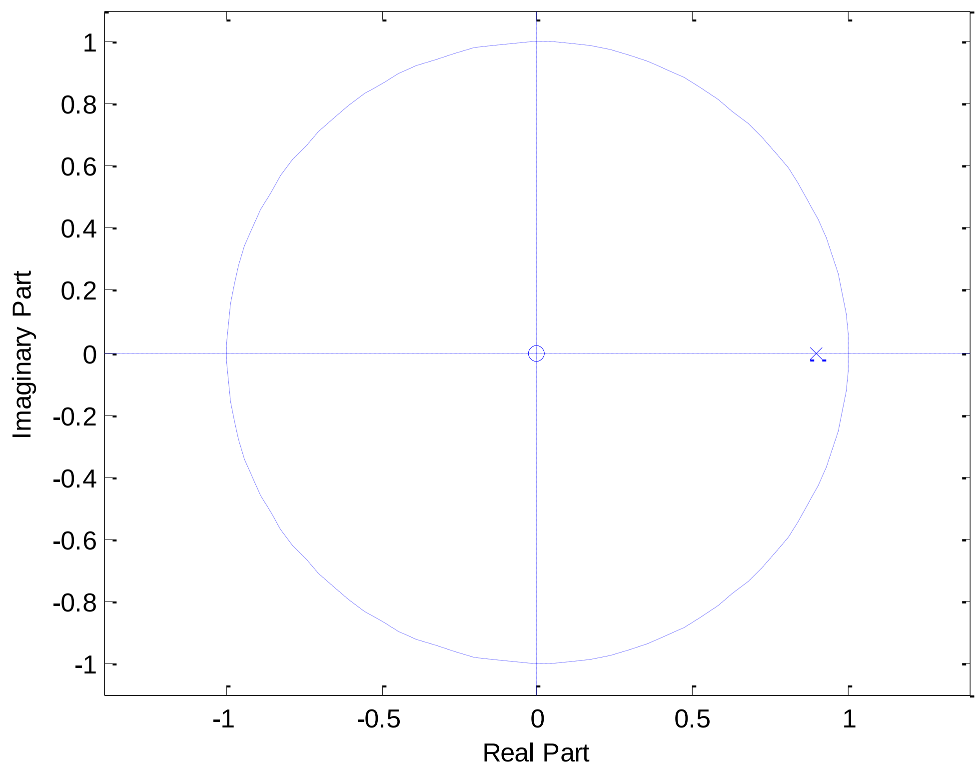
$$y(\mathrm{n}) - 0.9\,\mathrm{y}(\mathrm{n}-1) = \mathrm{x}(\mathrm{n})$$

$$\therefore H(\mathrm{z}) = \frac{1}{1-0.9\mathrm{z}^{-1}}$$

MATLAB Code:

```
b = [1, 0];
a = [1, -0.9];
zplane(b,a)
```

Output:

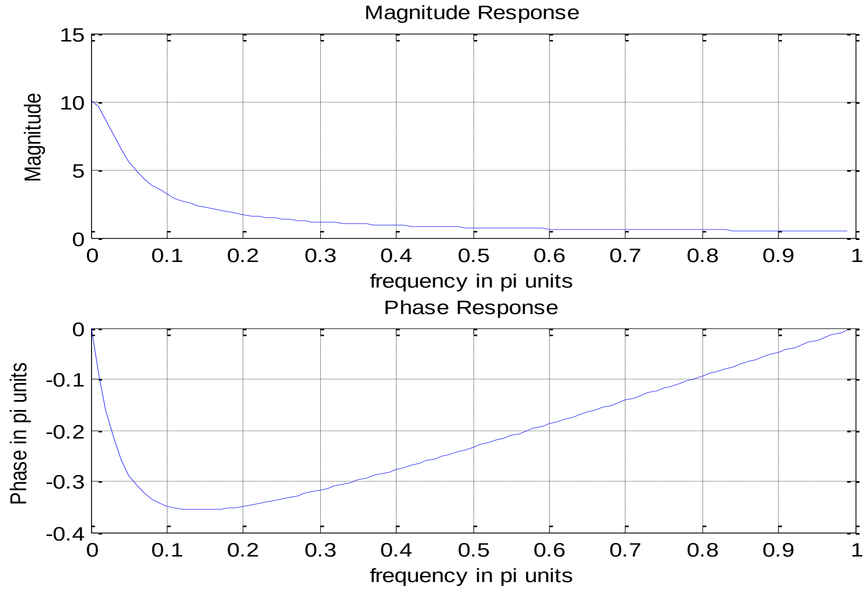


(b)

MATLAB Code:

```
[H,w] = freqz(b,a,100); magH = abs(H);
phaH = angle(H);
subplot(2,1,1);plot(w/pi,magH);grid
xlabel('frequency in pi units');
ylabel('Magnitude');
title('Magnitude Response')
subplot(2,1,2);plot(w/pi,phaH/pi);grid
xlabel('frequency in pi units');
ylabel('Phase in pi units');
title('Phase Response')
```

Output:



(c) **MATLAB Code:**

```
syms z H
H=z/(z-0.9);
h=iztrans(H)
```

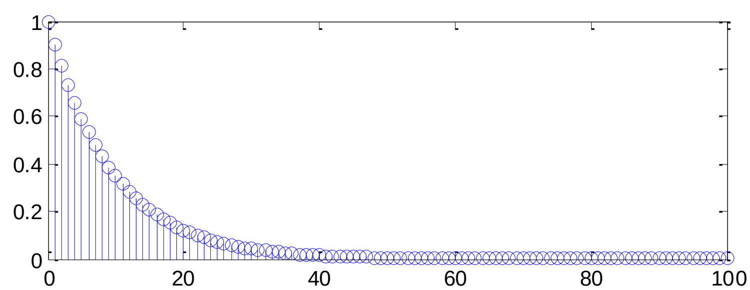
Output :

h =

(9/10)^n

Now we can plot it:

```
n=0:100
h = (9/10).^n;
stem(n,h)
```



Lab Session 4.5: Stability and Causality using z-Transform

Things to know:

Effect of poles on stability.

(i) **Poles outside the unit circle(|z|>1):** lead to exponential growth even if the input is bounded.

(ii) **Multiple Poles on the unit circle:** Always results in a polynomial growth.

(iii) **Simple (non-repeated) Poles on the unit circle :**Can lead to an unbounded response.

Effect of ROC on stability and causality.

The ROC of stable LTI system must include the unit circle

(i) **Stable causal system:** All the poles must lie inside the unit circle.

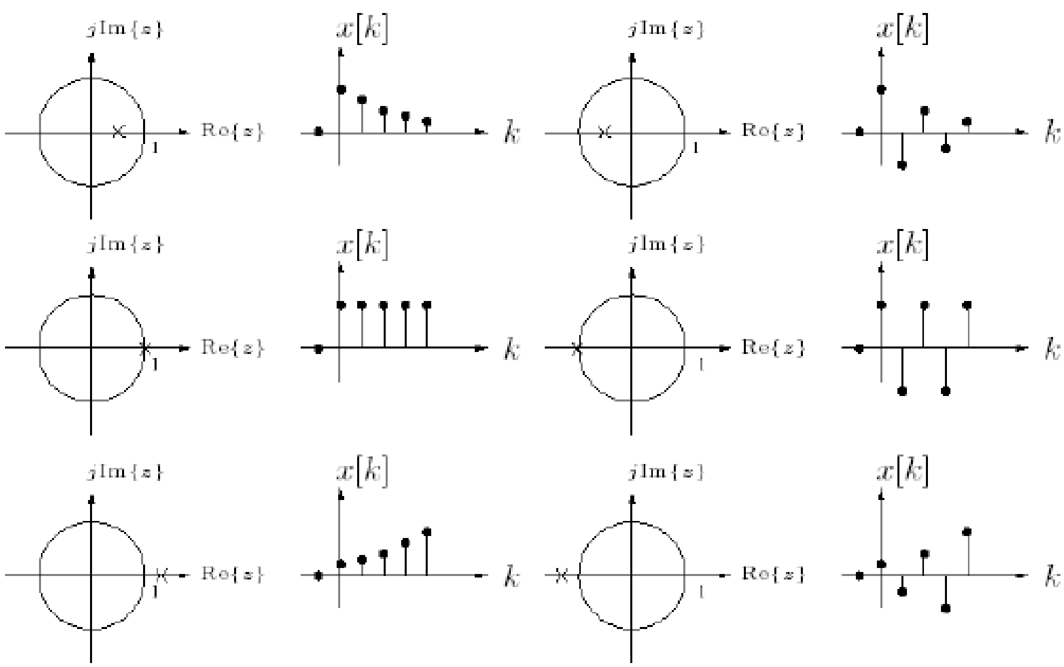
(ii) **Stable anti-causal system:** All the poles must lie outside the unit circle.

(iii) **Stability from Impulse Response:** h(n) must be absolutely summable.

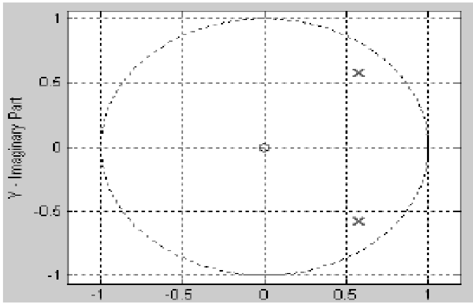
Graphical representation to show effect of pole locations on stability.

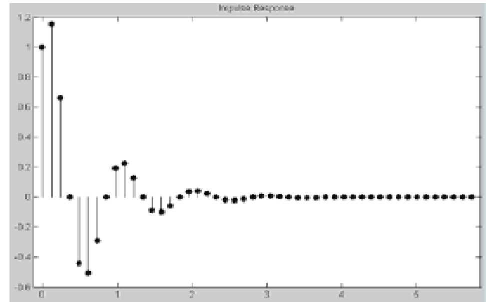
(i)Effect of real poles

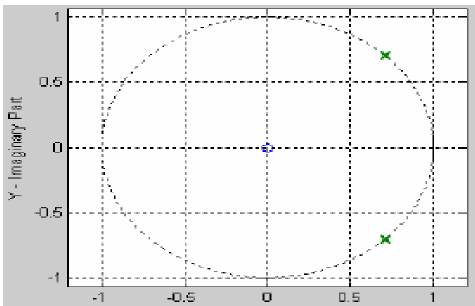
$$x[k] = a^k \epsilon[k] \circ\!\!-\!\!\bullet X(z) = \frac{z}{z-a} \qquad \text{ROC } |z| > |a|$$

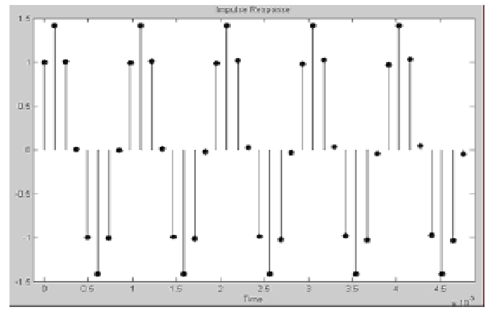


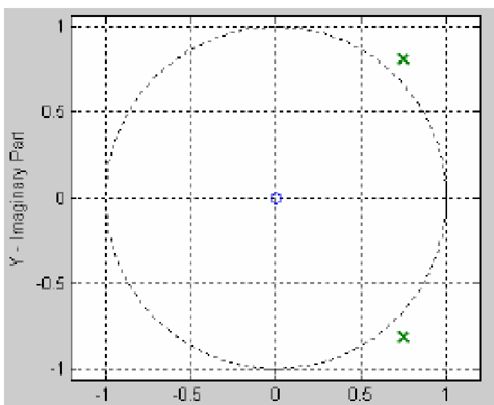
(ii)Effect of complex poles

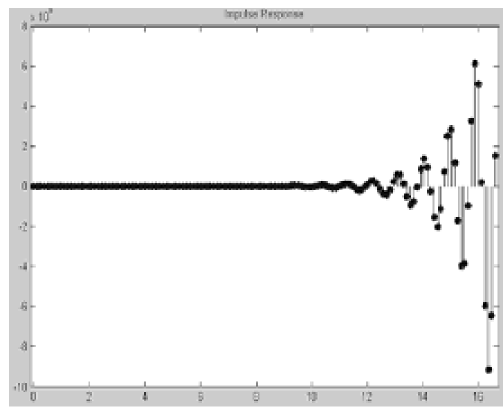




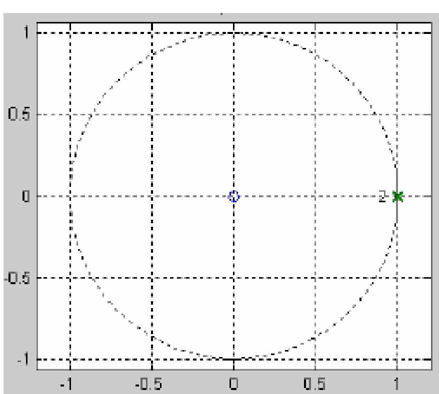


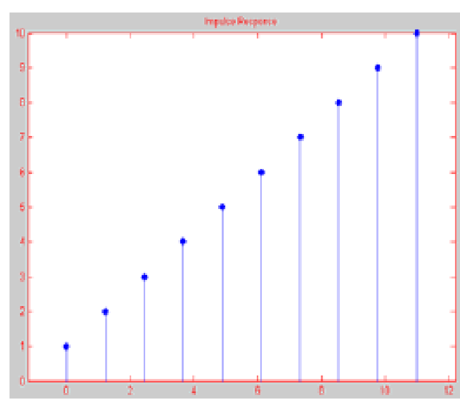






(iii)Effect of multiple real poles:





Matlab Example:

Example 4.6: Consider the system having transfer function,

$$H(z) = \frac{1+z^{-1}}{1-0.9z^{-1}+0.81z^{-2}}$$

(a) Sketch the impulse response using MATLAB. Comment on the stability of the system.

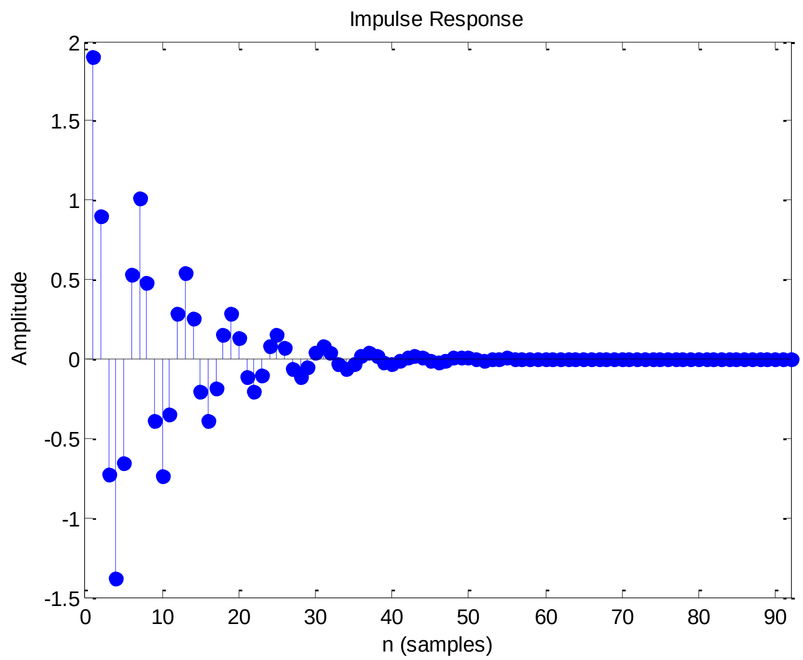
(b) Plot pole zero diagram and comment on the stability of the system assuming that the system is causal.

Solution:

(a) MATLAB code:

```
b=[1 1 0];
a=[1 -0.9 0.81];
impz(b,a);
h=impz(b,a);
sum(h)
```

Output:



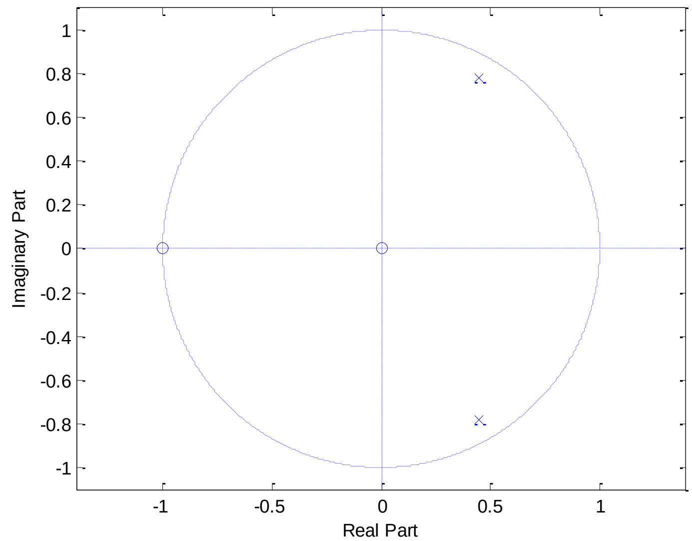
Numerical output at Command Window: ans = 2.1979

Comment: Since $\sum_{n=-\infty}^{\infty} |h(n)| < \infty$, the system is stable. It is also verified from figure.

(b) MATLAB code:

```
b=[1 1];
a=[1 -0.9 0.81];
zplane(b,a)
```

Output:



Comment: Poles lie inside the unit circle, so stable

Lab Session 4.6: Introduction to DTFT

If $x(n)$ is absolutely summable, that is, $\sum_{n=\infty}^{\infty} |\mathrm{x(n)}| < \infty$, then its discrete time Fourier transform is given by,

$$X(\mathrm{e}^{\mathrm{j}\omega}) \triangleq \sum_{n=\infty}^{\infty} \mathrm{x(n)}\,\mathrm{e}^{-\mathrm{j}\omega \mathrm{n}}$$

Function to compute DTFT

```
function [X] = dtft( x,n,w )
X=x*exp(-j*n'*w);
end
```

Example 4.7:For the following sequences, defermine DTFT $X(\mathrm{e}^{\mathrm{j}\omega})$. Plot the magnitude and angle of $X(\mathrm{e}^{\mathrm{j}\omega})$.

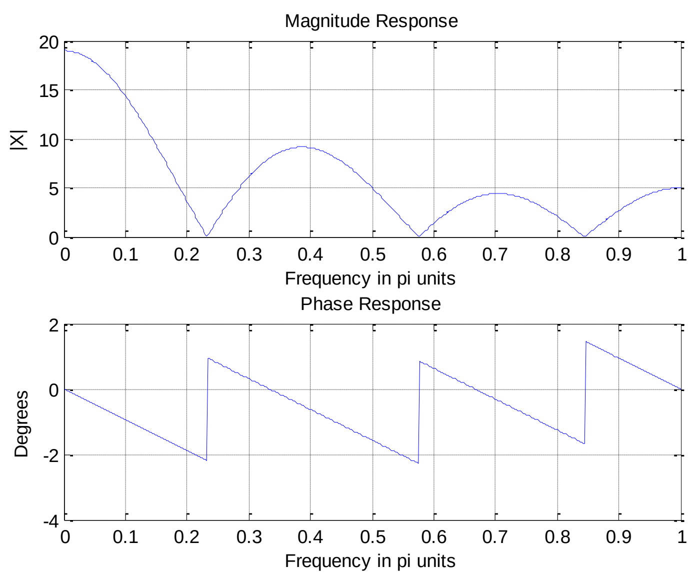
$$x(\mathrm{n}) = \{4, 3, 2, 1, 2, 3, 4\}$$

Solution:

MATLAB Code:

```
n=0:6;
x=[4 3 2 1 2 3 4];
w=[0:500]*pi/500;
X=dtft(x,n,w);
magX=abs(X);phaX=angle(X);
%Magnitude Response
subplot(2,1,1); plot(w/pi,magX);
grid;
xlabel('Frequency in pi units');
ylabel('|X|');
title('Magnitude Response');
%Phase Response
subplot(2,1,2);plot(w/pi,phaX);grid;
xlabel('Frequency in pi units');
```

Output:



Lab Session 4.7: In Lab Evaluation

ILE01 Consider the system having transfer function,

$$H(z) = \frac{1+2z^{-1}}{1-0.5z^{-1}+z^{-2}}$$

(i) Sketch the impulse response using MATLAB. Comment on the stability of the system.

(ii) Plot pole zero diagram and comment on the stability of the system assuming that the system is causal

ILE02 Compute and plot the frequency response of a causal LTI discrete-time system with a transfer function given by,

$$\text{(a)}\quad H(z) = \frac{0.15(1 - z^{-2})}{1 - 0.5z^{-1} + 0.7z^{-2}}$$

$$\text{(b)}\quad H(z) = \frac{0.15(1 - z^{-2})}{0.7 - 0.5z^{-1} + z^{-2}}$$

What is the difference between the two filters in (a) and (b), respectively? Which one will you choose for filtering and why?

Home Work :

Q1. Consider the system having transfer function,

$$H(z) = \frac{1 - 1.2z^{-1}}{1 - 0.5z^{-1}}$$

(i) Plot the impulse response using MATLAB. Comment on the stability of the system.

(ii) Plot pole zero diagram and comment on the stability of the system.

Q2. Consider the system having transfer function,

$$H(z) = \frac{1 + z^{-1}}{1 - 2z^{-1} + z^{-2}}$$

(i) Plot the impulse response using MATLAB. Comment on the stability of the system.

(ii) Plot pole zero diagram and comment on the stability of the system.

Q3. Consider a system which has poles at

$$p_{1,2} = re^{\pm j\pi/4}$$

and a zero at

$$z_1 = 0.707r$$

where: (i) r = 0.96 (ii) r = 0.71 and (iii) r = 0.14

(a) Derive analytically the impulse response of the system and show its dependence on r. Sketch the impulse response for r=0.96

(b) Plot the frequency and phase response for case (i)

(c) Plot the frequency and phase response for case (ii)

(d) Plot the frequency and phase response for case (iii)

(e) Observe the differences in the the three plots.

Q4. Consider the following system:

$$H(z) = \frac{z^{-3} - 1.8z^{-2} + 1.62z^{-1} - 0.729}{1 - 1.8z^{-1} + 1.62z^{-2} - 0.729z^{-3}}$$

(a) Find the poles and zeros of the transfer function.

(b) Plot the frequency and phase response of the system **on a linear scale**.

(c) Examine the frequency and phase response.

(d) What is the special characteristic of the system.